

OCTOBER 20 2022

Near-infrared photoglottography for measuring multiple glottal events

Yujie Chi  ; Kiyoshi Honda; Jianguo Wei



JASA Express Lett. 2, 105203 (2022)

<https://doi.org/10.1121/10.0014810>



LEARN MORE

Advance your science and career as a member of the
Acoustical Society of America

Near-infrared photoglottography for measuring multiple glottal events

Yujie Chi,^{a)}  Kiyoshi Honda, and Jianguo Wei^{b)}

College of Intelligence and Computing, Tianjin University, Tianjin, 300350, China

yujiechi@qq.com, khonda@sannet.ne.jp, jianguo@tju.edu.cn

Abstract: Near-infrared (NIR) photoglottography (PGG) is a non-invasive method for monitoring glottal activities which retains functionality of conventional PGG using visible light with more convenient accessibility. This paper is to investigate its performance in comparison with simultaneously recorded electroglottography (EGG) signals. Results showed that NIR PGG detects continuous transillumination for glottal aperture and vocal-fold contact. Glottal timing markers known as glottal closure and opening instants are detectable agreeing to the corresponding EGG-based instants. Further, it was inferred that variations of glottal waveforms based on NIR PGG reflect vertical vocal-fold edge motions. © 2022 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<http://creativecommons.org/licenses/by/4.0/>).

[Editor: Charles C. Church]

<https://doi.org/10.1121/10.0014810>

Received: 4 May 2022 **Accepted:** 22 September 2022 **Published Online:** 20 October 2022

1. Introduction

Glottographic detection of vocal-fold vibration is convenient for laboratory studies on speech physiology and novel applications in speech engineering. Photoglottography (PGG) has been commonly used for recording laryngeal articulation during speech as glottal abduction-adduction movements. For voice research, PGG has also been employed to monitor glottal aperture changes during vocal-fold vibration.^{1,2} PGG signals during voicing are characterized by the term *intermittent triangular*, i.e., cyclic repetitions of a steady-state baseline for glottal closure followed by a rise-and-fall curve clearly corresponding to glottal opening and closing periods. In contrast, electroglottography (EGG) detects changes in vocal-fold contact area, and EGG signals are less straightforward in interpreting glottal phases, often requiring their derivative waveforms to detect glottal closure and opening timings.³ Integrated analyses with the two devices have been reported as essential to obtain a complete picture of glottal activities, with EGG for the closed phase and PGG for the open phase.^{4,5} The transition points on recorded waveforms known as glottal closure instants (GCIs) and glottal opening instants (GOIs) could be identified by both methods.

Since the glottis is veiled inside the larynx, conventional PGG methods are moderately invasive requiring a light guide for illuminating the glottis passing through the nasal or oral cavity, which limits its applications in speech research. A non-invasive PGG technique, known as ePGG, was then developed with external lighting and sensing components.^{6–8} By adopting the NIR diagnostic window for biological tissues, a small near-infrared (NIR) light source was applied to obtain light transmission across the skin and muscles around the larynx. With the further improved illumination and amplification of noise reduction circuits, NIR PGG was successful at detecting glottal activities by our group⁹ and others.¹⁰ However, observed NIR PGG waveforms appeared more like asymmetrical triangular, lacking the steady-state glottal closure phase found in conventional PGG waveforms. See the supplementary material (Sec. 1) for a comparison between conventional PGG and NIR PGG waveforms.¹¹

The portable PGG (P-PGG) is our latest development for observing vocal-fold vibration by improving AC performance and signal stability.¹² Improvements with a higher-power IR LED and higher-gain amplifier revealed detailed glottal waveforms when high-pass filtering was applied to remove baseline fluctuation caused by speech articulation. In preliminary experiments, the use of P-PGG was successful for most of male and female subjects.

This letter is to further investigate the characteristics of NIR PGG glottal waveforms and discuss the relation between NIR light transmittance and tissue deformation of the vocal-fold mucous membrane. Audio, EGG, and P-PGG signals are recorded simultaneously, and temporal landmarks for glottal opening and closing events are compared.

^{a)} Author to whom correspondence should be addressed.

^{b)} Also at: Qinghai Nationalities University, Xining, 810007, China.

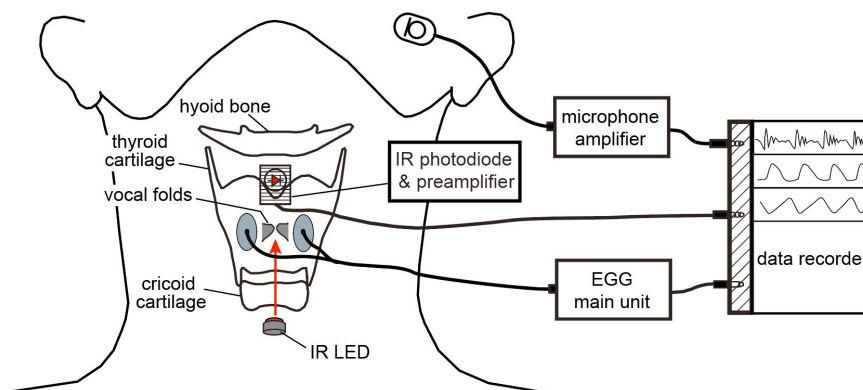


Fig. 1. Experimental setup for combined audio, EGG and NIR PGG recording. The portable PGG (P-PGG) employs NIR transillumination from bottom up along the midline of the neck. The EGG electrodes are placed with a common procedure.

2. Experimental and analysis methods

2.1 Equipment

A set of a microphone, EGG, and P-PGG was used in this study, as shown in Fig. 1. This setup is the same as in our report introducing P-PGG.¹² For P-PGG recording, an NIR LED (850 nm, 3 W) is placed below the cricoid cartilage, and a detector unit housing a PIN photodiode (BPW34) and transimpedance amplifier (60 M Ω) in a metal enclosure is placed above the larynx near the thyroid notch. This midline placement of P-PGG components is preferred for the ease with EGG measurement (EG2-PCX, Glottal Enterprises). The EGG electrodes are placed left and right at the glottis level. The multichannel EGG system is adequate for the experiment because it has no automatic gain control unit causing extra time delay and amplitude distortion.¹³ Together with audio signals from a microphone (AT9903, Audio Technica), all the signals are recorded into a multichannel data recorder (DAS 40, Sefram). For the purpose of temporal comparisons between P-PGG and EGG signals, the primary output of the P-PGG preamplifier (DC $\sim$ 5 kHz bandwidth) was directly sampled by the recorder and processed by *post hoc* digital bandpass filtering to avoid time delay.

2.2 Experiment

Four healthy Chinese speakers aged 24 to 30 years old (two males ZZ, ZX, and two females CY, XY) were engaged in this experiment. The speakers wore the EGG electrodes and microphone first. Then, they were instructed to hold the light emitter and detector units pressed onto the midline neck skin separately. The experimenter helped adjust the positions and ensured effective recordings on a signal monitor. Finally, vowels (/a/, /i/, and /u/) were produced three times with normal voice and flat tone. Audio, EGG, and P-PGG signals were recorded at a sampling rate of 50 kHz. All the experiments were conducted in a soundproof room.

2.3 Signal preprocessing

The raw P-PGG signals were processed with PYTHON toolkits. To reduce signal fluctuations and high-frequency noise, a bandpass filter of 30 Hz to 3 kHz was adopted. To do so, a third-order Bessel filter (scipy.signal.bessel) and zero-phase digital filtering (scipy.signal.filtfilt) were applied to minimize phase errors. EGG signals were also processed by a high-pass filter of a 30-Hz cutoff to reduce fluctuations. Table 1 introduces the signal-to-noise ratio (SNR) of the P-PGG and EGG signals for each speaker. Filtering on the P-PGG signals improved the SNR about by 8 dB. The higher SNRs for EGG signals were attained by male speakers than by female speakers. The male speaker ZX exhibited the lowest SNR for P-PGG.

The simultaneous recordings of audio, EGG and P-PGG are presented in Fig. 2 in panels A–D for four speakers. The EGG signals were inverted to show vocal-fold contacts as negative deflections in accordance with glottal closing in P-PGG signals. The derivative EGG signals (dEGG in gray) were drawn along with EGG curves. Audio signals were shifted

Table 1. Quality of glottographic signals for four speakers.

Speaker	Gender	Mean f_0 (Hz)	SNR (EGG) (dB)	SNR (P-PGG/filtered) (dB/dB)
ZZ	Male	118	51	24/32
ZX	Male	135	48	15/24
CY	Female	219	42	22/29
XY	Female	244	33	19/27

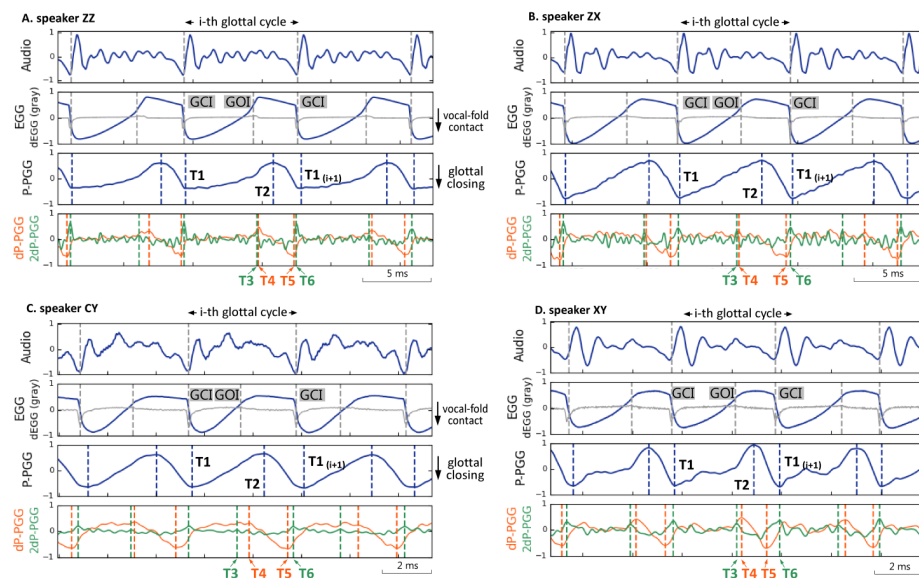


Fig. 2. Amplitude normalized waveforms of audio, inverted EGG, dEGG (gray), P-PGG, dP-PGG (orange), and 2dP-PGG (green) signals for four speakers (panels A–D). Each segment includes three glottal cycles of vowel/a/. The dashed vertical lines T1–T6 highlight landmarks on the glottal cycle: T1, moment of P-PGG trough; T2, P-PGG peak; T4, dP-PGG peak; T3, 2dP-PGG peak shortly before T4; T5, dP-PGG trough; and T6, 2dP-PGG peak shortly after T5.

left in time to compensate for sound propagation time from the glottis to the microphone. The delay time applied was 0.59 ms for male speakers according to the estimates of their vocal-tract length of 17.5 cm with an additional 2.5-cm distance from the lips to the microphone. For females, the delay time was estimated at 0.53 and 0.45 ms for CY and XY, respectively. All the waveforms were normalized for amplitude, and then segments of vowel /a/ were enlarged for showing three cycles. See the supplementary material (Sec. 2) for additional data of vowels /i/ and /u/.¹¹

2.4 Comparing glottal events of P-PGG and EGG signals

On P-PGG signals, landmarks T1 and T2 (blue dashed line) were identified to locate P-PGG peaks and troughs, respectively. T1 is labeled on the trough, as the starting point of the i -th glottal cycle, and T2 on the peak within the cycle. The P-PGG signals show less obvious variations during glottal closing and opening periods. To identify those glottal events, the first and second derivatives of the filtered P-PGG signals were computed as shown in Fig. 2 (bottom) by dP-PGG (in orange) and 2dP-PGG (in green), respectively. On dP-PGG signals, landmarks T4 and T5 (orange dashed lines) are labeled at the moments of the maximum dP-PGG peak and minimum trough, respectively. Subsequently, on the 2dP-PGG signals, the landmark T3 (green dashed lines) is a peak shortly before T4, and T6 is a peak after T5. Finally, $T1_{(i+1)}$ is the initiation of the next glottal cycle.

For further comparisons, the moments on extreme dEGG values were marked (dashed gray lines) for glottal closure instants (GCIs) and glottal opening instants (GOIs). Occasionally, there were a few candidates for the dEGG trough, and the leftmost one was adopted for quantitative analysis in this study. In addition, a five-point moving average was applied for finding the dEGG peaks within a glottal cycle.

The temporal relationship between the landmarks on P-PGG and EGG signals was then examined statistically. For each glottal cycle, $T1_{(i+1)}$, T5, and T6 were compared with the reference GCIs. T3 and T4 were compared with the reference GOIs. Finally, the results obtained from all the cycles were depicted in time-interval histograms.

3. Results

A gross observation of the data in Fig. 2 reveals that P-PGG signals differ from conventional PGG signals for normal voice. As a natural interpretation, P-PGG maintains the merit of tracing glottal closing ($T2-T1_{(i+1)}$), while indicating a rising pattern from the initial to maximum opening ($T1-T2$). Across four speakers, the signal shapes on rising slopes are quite different. For speaker ZZ in Fig. 2(A), the P-PGG curve resembles the known glottal area waveform with a gradual rise of the baseline in the closed period. For other two speakers in Figs. 2(B) and 2(C), the rising slopes appear linear without distinct inflection points, agreeing with the corresponding EGG curves showing the smoother changes near the GOIs. For speaker XY in Fig. 2(D), the P-PGG curve exhibits a ripple during the closed period.

From the above observations, P-PGG signal characteristics are questioned for their causal process, and quantitative comparisons between P-PGG and EGG signals are further conducted. In what follows, temporal relationships between

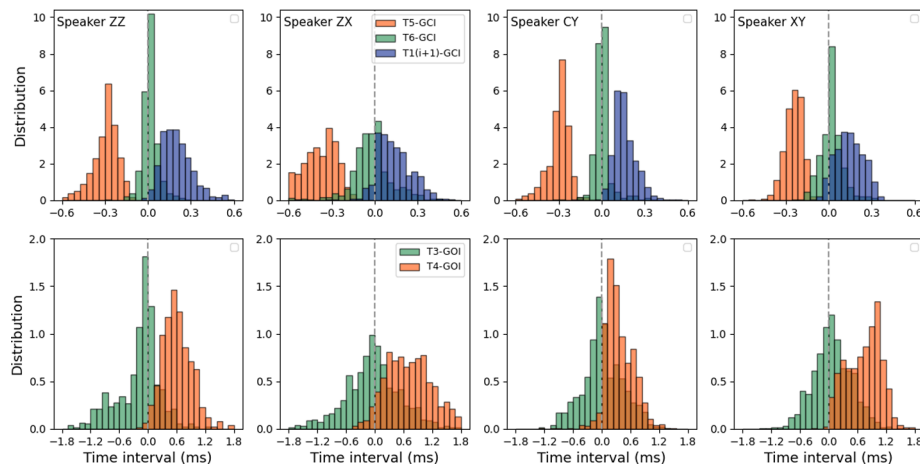


Fig. 3. Histograms of time intervals from P-PGG landmarks to EGG-GCIs (top row) and EGG-GOIs (bottom row).

landmarks on P-PGG curves (T1–T6) and those on EGG signals (GCI and GOI) were examined by computing time-interval histograms. The results are presented in Fig. 3.

3.1 Glottal closure timing

The top row of Fig. 3 shows three time-interval histograms in each panel for each speaker. The intervals were computed for the periods from three P-PGG landmarks, T1(i + 1), T5, and T6, to the GCIs on the EGG signals. The bar colors are consistent with the colors of P-PGG, dP-PGG, and 2dP-PGG waveforms in Fig. 2. It is obvious from the three histograms in each panel that the moment of the 2dP-PGG peak (T6) is located closer to GCIs, implying a good estimate of GCIs. The estimation errors are within ± 0.1 ms for P-PGG signals of good quality (for speakers ZZ, CY, and XY).

The results for all four speakers show that P-PGG signals demonstrate a consistent temporal pattern as observed in conventional PGG signals regarding the closing timing. The P-PGG troughs (T1) maintain about 0.1-ms lagging and errors of ± 0.2 ms to GCIs. This suggests that NIR transillumination reaches the minimum immediately after a vocal-fold contact at the lower edges.

3.2 Glottal opening timing

The bottom row of Fig. 3 presents the histograms of time intervals from landmarks T3 and T4 to GOIs on EGG (T4 for the dP-PGG peak and T3 for 2dP-PGG peak before T4). For speaker ZZ, T3 shows the best approximate of the glottal opening events. The error range is about ± 0.3 ms, three times greater than the estimates of GCIs. This is reasonable because vocal-fold de-contact is more gradual than the contact. Even though the dP-PGG peaks (T4) for speaker ZX and CY are not prominent in Fig. 2, the landmark T3 also shows a visible correspondence with the GOIs. For speaker XY, the histogram shows two clusters for the landmark T4, while T3 still indicates the good approximate of the GOIs. The cause for this variation will be analyzed in Sec. 3.3.

3.3 Double-peaked EGG derivative (dEGG) for glottal opening

Through the whole observation on the dEGG curves, double positive peaks are found frequently for speaker XY. It means that there are two candidates for the GOI within a glottal cycle. Accordingly, two candidate peaks were sought on the 2dP-PGG curves having the similar timing.

Figure 4(a) depicts the case showing double peaks both on dEGG and 2dP-PGG from speaker XY. The two dEGG peaks are marked as E1 and E2, and the two 2dP-PGG peaks are marked as P1 and P2. The average interval between E1 and E2 is about 0.5 ms. The time-interval histograms for P1 to E1 (-0.5 ms offset) and P2 to E2 are drawn in Fig. 4(b).

The above results suggest the relevance of 2dP-PGG to dEGG in identifying glottal opening events. As has been reported from ultra-high-speed video recordings of vocal-fold vibration, the double peaks in dEGG seem to occur due to the anterior-posterior phase difference.¹⁴ If this is the case, it can be inferred that P-PGG is sensitive to the glottal transillumination derived from longitudinal deformation of the vocal folds, although it is a speculation requiring further investigations.

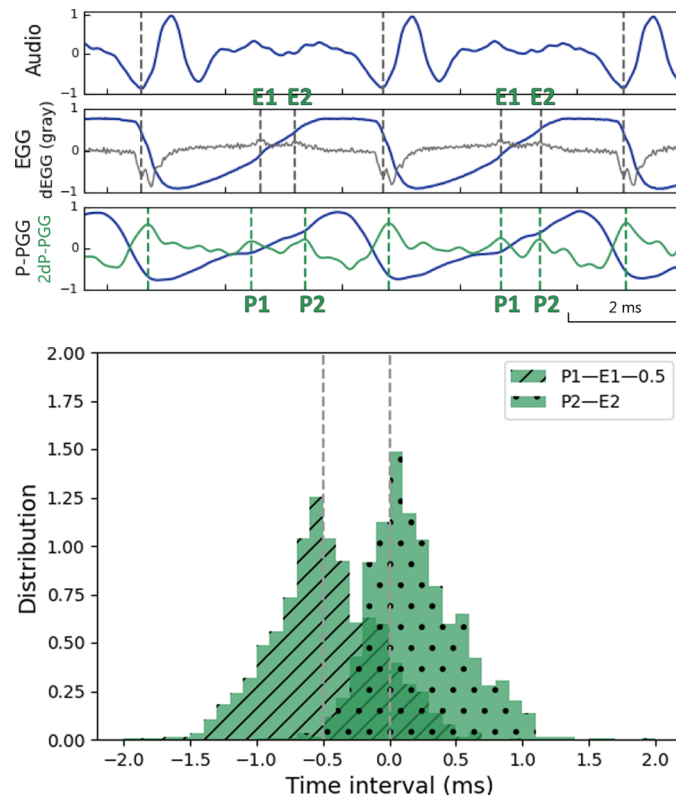


Fig. 4. A case of double peaks on dEGG and 2dP-PGG. (a) Landmarks of glottal opening events for EGG and P-PGG waveforms from speaker XY: E1, moment of dEGG peak in the left; E2, dEGG peak in the right; P1, moment of 2dP-PGG peak in the left of P2; and P2, 2dP-PGG landmark T3. (b) Two time-interval histograms from P1 to E1 (−0.5 ms offset, hatched) and P2 to E2 (dotted).

4. Discussions and conclusion

In this letter, a new portable photoglottography (P-PGG) was introduced as a non-invasive system for detecting multiple glottal events during vocal-fold vibration. Besides indicating the glottal maximum closure and peak opening, P-PGG signals are interpretable to determine the glottal closure and open instants (GCI, GOI) jointly with temporal analyses on electroglottographic (EGG) recordings. Two peaks on the second derivatives of P-PGG (2dP-PGG) were identified to indicate the GCI and GOI. As for glottal closing, the late peak (T6) of 2dP-PGG agrees with GCI on EGG within the error of ± 0.1 ms. For glottal opening, P-PGG signals gradually rise but still show a relationship between the early peak (T3) on the 2dP-PGG and GOI on EGG. On this basis, P-PGG landmarks T1–T6 are considered as reliable timings of glottal events, possibly involving its lower and upper edges.

The present P-PGG is one of the near-infrared photoglottographic (NIR PGG) systems, and its signals differ from those of conventional visible-light PGG showing stable glottal closure periods. NIR PGG signals are thought to reflect both changes in glottal aperture and vocal-fold contact partly because the vocal-fold tissues are semi-transparent to NIR light. The results of this study comparing P-PGG and EGG signals infer vocal-fold vibration patterns with certain further details. At glottal closing, P-PGG signals slope down fast and reach the minima of transillumination at the initiation of glottal closure, which agrees with the slope of conventional PGG.¹⁵ At glottal opening, P-PGG signals show a gradual rise with occasional ripples and reach approximately half of the amplitude at the opening moments (EGG-GOIs). It is speculated that movements of the vocal-fold lower edges are dominant in deriving P-PGG signal variations, and those of the upper edges only partially account for the glottal opening. One possible explanation is that the upper part of the vocal folds contains more mucous and connective tissues,¹⁶ which have the higher optical transmissivity than the inner vocalis muscle.¹⁷ The transmittance of the upper edge will also be larger for edge sharpening at opening by thinning vertically and widening horizontally.¹⁸ Assuming a total transmittance index of 50% for the upper edge and 0% for the lower edge, the equivalent area (A) from NIR transillumination will be

$$A = \begin{cases} A_{\text{lower}} + (A_{\text{upper}} - A_{\text{lower}}) \times 0\%, & A_{\text{upper}} \geq A_{\text{lower}}, \\ A_{\text{upper}} + (A_{\text{lower}} - A_{\text{upper}}) \times 50\%, & A_{\text{lower}} > A_{\text{upper}}, \end{cases} \quad (1)$$

where A_{upper} represents the area between the upper edges and A_{lower} the area between the lower edges. The transmittance indices may vary across different speakers and voice types, e.g., smaller for the upper edges of speakers CY and ZX in this

study. For conventional PGG, the transmittance indices are ideally zero for visible light, and the glottal area A equals the smaller of A_{upper} and A_{lower} during glottal cycles.

The occasional double peaks found on EGG derivatives at glottal opening have been explained by ultra-high-speed vocal-fold imaging, which suggested certain phase differences during longitudinal de-contacting.¹⁴ In contrast to such two-dimensional observations, however, an alternative account can be proposed from three-dimensional MRI of the vocal folds as shown in the supplementary material (Sec. 3) for the vertical extensions of the vocal-fold edges.^{11,19} During phonation, the vocal-fold edges show vertical extensions of the mucous membrane near the vocal processes. This vertical mass may exhibit coupled oscillations with the rest of the edges resulting in delayed separation of the portions. The ripples on P-PGG signals at opening may also reflect such a complexity during a glottal opening sequence.

The NIR PPG has been advanced from the earlier systems by improved lighting and sensing components to be capable of capturing transillumination during glottal cycles, while it still maintains certain limitations, such as signal variations due to speaker's neck tissue geometry. Further improvements on the system and other methods to combine would be necessary to clarify the details of vocal-fold vibration and evaluate voices of various types.

Acknowledgments

This work was supported by National Key R&D Program of China (Grant No. 2020YFC2004103), National Natural Science Foundation of China (Grant Nos. 61876131 and U1936102).

References and lists

- ¹T. Baer, A. Löfqvist, and N. S. McGarr, "Laryngeal vibrations: A comparison between high-speed filming and glottographic techniques," *J. Acoust. Soc. Am.* **73**(4), 1304–1308 (1983).
- ²W. Habermann, J. Jiang, E. Lin, and D. G. Hanson, "Correlation between glottal area and photoglottographic signal in normal subjects," *Acta Oto-laryngol.* **120**(6), 778–782 (2000).
- ³C. T. Herbst, "Electroglottography—An Update," *J. Voice* **34**(4), 503–526 (2020).
- ⁴J. J. Jiang, S. Tang, M. Dalal, C-h Wu, and D. G. Hanson, "Integrated analyzer and classifier of glottographic signals," *IEEE Trans. Rehabil. Eng.* **6**(2), 227–234 (1998).
- ⁵B. R. Gerratt, D. G. Hanson, and G. S. Berke, "Laryngeal configuration associated with glottography," *Am. J. Otolaryngol.* **9**(4), 173–179 (1988).
- ⁶K. Honda and S. Maeda, "Glottal opening and airflow pattern during production of voiceless fricatives: A new non-invasive instrumentation," *J. Acoust. Soc. Am.* **123**(5), 3738 (2008).
- ⁷K. Honda and S. Maeda, "Non-invasive photoelectro-glottography method and device," U.S. patent 20100256503A1 (October 7, 2010).
- ⁸J. Vaissière, K. Honda, A. Amelot, S. Maeda, and L. Crevier-Buchman, "Multisensor platform for speech physiology research in a phonetics laboratory," *J. Phon. Soc. Jpn.* **14**(2), 65–77 (2010).
- ⁹Y. Chi, K. Honda, and J. Wei, "Glottographic and aerodynamic analysis on consonant aspiration and onset F0 in Mandarin Chinese," in *ICASSP 2019—2019 IEEE International Conference on Acoustics, Speech and Signal Processing* (2019), pp. 6480–6484.
- ¹⁰E. Suthau, P. Birkholz, A. Mainka, and A. P. Simpson, "Non-invasive photoglottography for use in the lab and the field," in *Proceedings of the 12th ITG Symposium on Speech Communication*, Paderborn, Germany (October 5–7, 2016), pp. 1–5.
- ¹¹See supplementary material <https://www.scitation.org/doi/suppl/10.1121/10.0014810> for additional data.
- ¹²Y. Chi, K. Honda, and J. Wei, "Portable photoglottography for monitoring vocal-fold vibrations in speech production," in *ICASSP 2021—2021 IEEE International Conference on Acoustics, Speech and Signal Processing* (2021), pp. 6438–6442.
- ¹³M. Rothenberg, "A multichannel electroglottograph," *J. Voice* **6**(1), 36–43 (1992).
- ¹⁴C. T. Herbst, J. Lohscheller, J. G. Švec, N. Henrich, G. Weissengruber, and W. T. Fitch, "Glottal opening and closing events investigated by electroglottography and super-high-speed video recordings," *J. Exp. Biol.* **217**(6), 955–963 (2014).
- ¹⁵D. G. Hanson, J. Jiang, M. D'Agostino, and G. Herzon, "Clinical measurement of mucosal wave velocity using simultaneous photoglottography and laryngostroboscopy," *Ann. Otol. Rhinol. Laryngol.* **104**(5), 340–349 (1995).
- ¹⁶M. Hirano, "Vocal mechanisms in singing: Laryngological and phoniatric aspects," *J. Voice* **2**(1), 51–69 (1988).
- ¹⁷J. Mobley, T. Vo-Dinh, and V. V. Tuchin, "Optical properties of tissue," in *Biomedical Photonics Handbook*, 2nd ed., edited by T. Vo-Dinh (CRC Press, Boca Raton, FL, 2014), Vol. 1, pp. 23–122.
- ¹⁸K. Ishizaka and J. L. Flanagan, "Synthesis of voiced sounds from a two-mass model of the vocal cords," *Bell Syst. Tech. J.* **51**(6), 1233–1268 (1972).
- ¹⁹K. Honda and J. Zhang, "Articulatory modeling of human and non-human vocal production," *J. Acoust. Soc. Am.* **143**, 1787 (2018).